

SAM SEIFI

svsseifi@gmail.com | github.com/svsam | linkedin.com/in/svssams | svsam.com

EDUCATION

BSc Physics with Astrophysics and Cosmology, King's College London

Sep 2025 - expected 2028

- **Year 1 average: 2:1**

Modules: Computing 91%, Laboratory 72%, Astrophysics 67%, Mathematics 65%.

GCE A-levels, The Compton Sixth Form

Sep 2023 - Aug 2025

- **Grades:** Mathematics (A), Physics (B), Computer Science (B).

RESEARCH AND TECHNICAL PROJECTS | SVSAM.COM

EvoXRB: Genetic Algorithms and a Very Real Black Hole

- Developed **EvoXRB**, a reproducible X-ray spectral-timing analysis pipeline combining **genetic optimisation**, local refinement, posterior inference and parameter-recovery testing. **Achieved 100% convergence** across **60 polished synthetic fits** and **98.95% mean parameter-recovery accuracy across 240 estimates**. Project write-up.

Exoplanet Orbital Temperatures and Eccentricity Modelling

- Processed **149,648 orbital-quarter records** into a deduplicated catalogue of **6,284 exoplanet–host systems**, resolving competing and incomplete parameters for consistent two-body analysis.
- **6,284 selectable orbits**, each supported by a **361-sample full-period** Keplerian pipeline calculating planet–star distance, vis-viva speed and equilibrium-temperature variation.

Earthaccess Contributor

- Contributed documentation to the open-source Python library earthaccess, which searches for and downloads or streams NASA Earth science data with only a few lines of code.

ASCII Art Generator

- For local, cross-platform image-to-text conversion, engineered a shared Rust/WebAssembly core with background workers, dithering and 24-bit ANSI export; benchmarked at **13.3 frame-equivalent conversions/s** and **approximately 96,000 output cells/s** for 512×512 inputs at 120 columns.

Minesweeper AI Solver

- Built a constraint-based solver that converts revealed cells into linear systems, enumerates valid mine placements, applies overlapping-set deductions and selects minimum-risk moves while identifying forced 50/50 states.
- Achieved a **73% win rate across 100 seeded 10×10 boards with 20 mines**, completing **941 probability calculations** with zero solver errors and detecting **21 forced 50/50 positions**.

Terraria Mod: Daybreak Class Extension

- Architected a C# 12 combat system around a network-synchronised **seven-state projectile machine** and data-driven weapon profiles, supporting 10 progression weapons across single-player, host-and-play and dedicated servers.
- Attracting **84 users** while maintaining static validation for balance, assets, packaging and multiplayer-sensitive logic.

EXPERIENCE

The Deli Downstairs

Jul 2026 - Present

Kitchen Porter, part-time

London, UK

- Supported high-volume kitchen service while maintaining clean, organised and hygienic preparation and equipment areas.

National Student Space Conference (NSSC)

Feb 2026

Volunteer

London, UK

- Volunteered at the conference hosted by King's College London, representing the university's space society and coordinating weekend events and specialist talks.

MedMart Borehamwood

Jul 2023 - Sep 2024

Pharmacist Assistant, part-time

London, UK

- Sorted and handled medication for elderly patients while coordinating sensitive, time-critical tasks with colleagues at over 90% accuracy.

LRC, The Compton School

Oct 2020 - Jun 2023

Volunteer Librarian

London, UK

- Assisted with library organisation and rentals, student enquiries and maintaining an orderly learning environment.

TECHNICAL SKILLS

Programming and tools: Python, Rust, C#, JavaScript, HTML/CSS, SQL, Linux, $\text{\LaTeX}$, pandas, APIs, Git, Astroquery, WebAssembly, Jupyter, Blender.

Laboratory: Oscilloscope, function generator, digital multimeter.